

## 1. Combine Two Tables

**TABLE: PERSON**

| personId | lastName | firstName |
|----------|----------|-----------|
| 1        | Wang     | Allen     |
| 2        | Alice    | Bob       |

**TABLE: ADDRESS**

| addressId | personId | city          | state      |
|-----------|----------|---------------|------------|
| 1         | 2        | New York City | New York   |
| 2         | 3        | Leetcode      | California |

### QUESTION

Write a SQL query to report the first name, last name, city, and state of each person in the Person table.

#### Rules:

- If the address of a personId is not present in the Address table, report null instead.
- Return the result table in any order.

### EXPECTED OUTPUT

| firstName | lastName | city          | state    |
|-----------|----------|---------------|----------|
| Allen     | Wang     | Null          | Null     |
| Bob       | Alice    | New York City | New York |

1) output

Select

p. firstName,

p. lastName,

a. city,

a. state

from person p

Left Join Address a

on p.personId = a.personId;

## 2. Employees Earning More Than Their Managers

TABLE: EMPLOYEE

| id | name  | salary | managerId |
|----|-------|--------|-----------|
| 1  | Joe   | 70000  | 3         |
| 2  | Henry | 80000  | 4         |
| 3  | Sam   | 60000  | Null      |
| 4  | Max   | 90000  | Null      |

### QUESTION

Write an SQL query to find the employees who earn more than their managers.

#### Rules:

- managerId is the id of the employee's manager.
- Return the result table in any order.

### EXPECTED OUTPUT

| Employee |
|----------|
| Joe      |

2) Select

e. name AS Employee  
from employee e  
join Employee m

on e.managerId = m.id  
where e.salary > m.salary;

**TABLE: PERSON**

| id | email   |
|----|---------|
| 1  | a@b.com |
| 2  | c@d.com |
| 3  | a@b.com |

**QUESTION**

Write an SQL query to report all the duplicate emails.

**Rules:**

- Emails will only consist of lowercase letters.
- Return the result table in any order.

**EXPECTED OUTPUT**

| Email   |
|---------|
| a@b.com |

3) select email AS Email (

from Person

Group by email

having count(email) > 1;